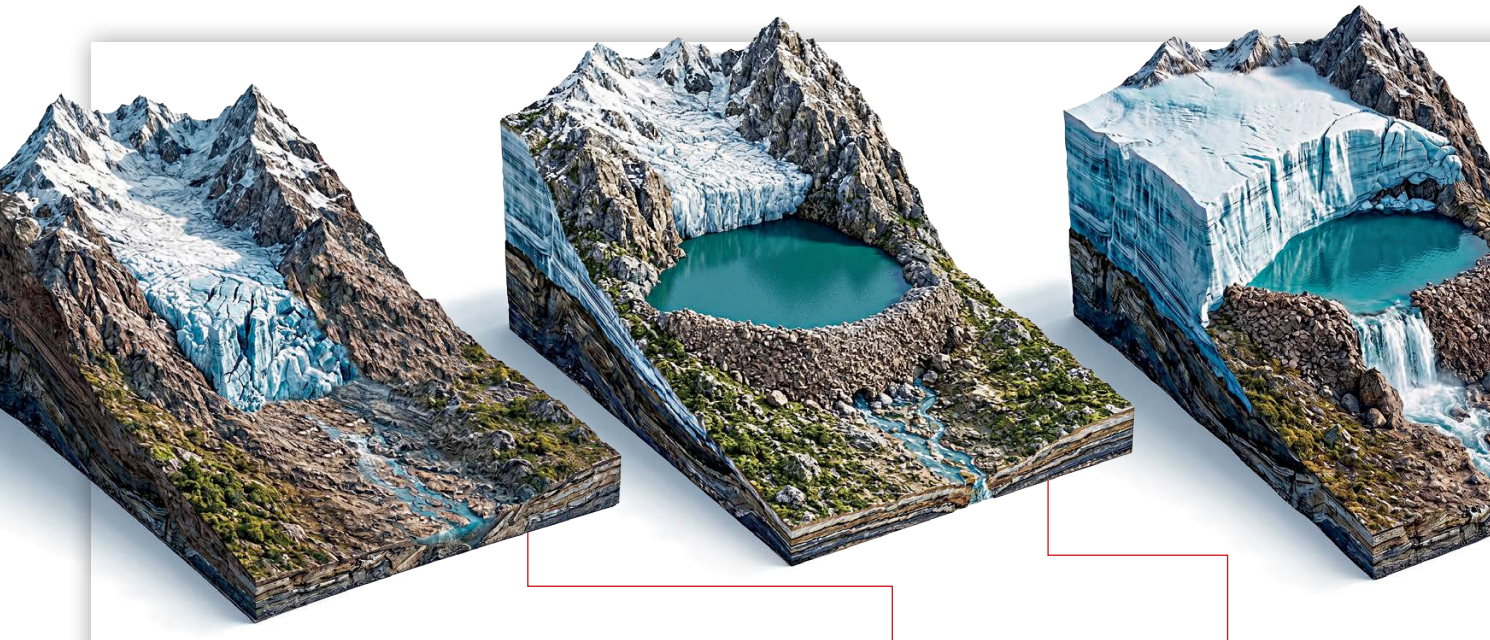


WHAT IS A GLOF?

Glacial Lake Outburst Flood or GLOF is the sudden release of a large volume of water from a lake formed by melting glaciers, usually when the natural dam holding the lake back fails. Once it gives way, water, ice, boulders and mud can rush down a mountain valley at enormous speed

Graphic by **NILANJAN DAS**

GLACIER MELTS AND RETREATS
Global warming causes the glacier to shrink, leaving a depression at its front, often visible from satellites

A GLACIAL LAKE FORMS AND GROWS
Meltwater is contained by a dam made of loose rock and debris called moraine or sometimes glacier ice itself. As melting continues, the lake becomes larger and deeper

earth. A week later, Nepal's disaster authority put the death toll at over 1,200, with nearly 5,000 still missing, among them pilgrims, trekkers and hydropower workers. Also, more than 200 Indians, most of whom were on a pilgrimage to Mount Kailash in Tibet.

Geologist Dr Umesh Haritashya, a professor of Earth, Environment and Sustainability at the University of Dayton, Ohio, US, who has done extensive research on glacier hydrology and high mountain hazards, describes the trigger with telling precision. "This was not a typical ice avalanche where ice just falls on the ground—it was a bedrock detachment, and ice happened to be sitting on that bedrock." The precise volume of rock and ice is still being calculated, and the source of all the water being tracked. But the physical sequence is now clear enough: a mountainside fell, a destructive debris flood raced through the valley, and temporary barrier lakes created by the wreckage posed the danger of further surges.

That distinction matters profoundly to India. The familiar phrase 'glacial lake outburst flood', or GLOF, conjures a lake swelling behind a natural dam of loose rock and sediment until that dam gives way. Nepal revealed a wider and more unpredictable family of threats: hanging glaciers can detach; thawing rock faces can collapse; an avalanche can plunge into a lake and throw water over its rim; debris can block a river, create a lake in hours and unleash a second flood when the obstruction fails. The lethal lake may not exist on any official map before the disaster begins. And the political boundary offers no protection. "Whatever happened there can happen in India," Haritashya warns. "Political boundaries really don't exist from a mountain-stability perspective."

THE HIGH RISK

Haritashya, who earned his doctorate from IIT Roorkee, is unfortunately very right. Even as the Indian government moved in to help Nepal with vital medical and food supplies in addition to rescue teams, it was apparent that the cause and the magnitude of the tragedy had serious implications for India's vast Himalayan region. The mountainous arc of 2,500 km stretches from Ladakh and Jammu & Kashmir through Himachal Pradesh and Uttarakhand to Sikkim and Arunachal Pradesh. In all, it spans 11 states and two Union territories, with an overall population of nearly 50 million. But the downstream population living in the river basins of the Ganga, Brahmaputra and Indus that could be impacted by Himalayan flooding is estimated to be more than 700 million, or half the country's total.

Moreover, much of the peril lurks far beyond India's borders because the headwaters of the Indus, Sutlej, Brahmaputra and several tributaries of the Ganga lie in Tibet, Nepal or Bhutan. A collapse in another country can become a disaster for India downstream. The scale is easy to underestimate. The Hindu Kush Himalaya (HKH) contains more than 63,000 glaciers. According to the Indian Space Research Organisa-